HINAKU · ヒナク

UI/UX Design System & Desktop Architecture

Specification, Component Hierarchies & Visual Design Tokens

An exhaustive design specification for HINAKU: a personal, local-first AI assistant built for one person. Grounded in the core brand aesthetic—*"A universe without light"* (光なき宇宙)—this document details the 3-column collapsible desktop shell, floating interaction composer, truthful telemetry accordions, document attachment walkthrough engine, semantic memory drawers, and complete accessibility standards.

TARGET ENVIRONMENT

HP V15 Laptop (AMD Ryzen 7, RTX 4050 6 GB VRAM)

SYSTEM ARCHITECTURE

FastAPI Backend + PyWebView Native Shell + Local Web UI

MODEL BACKENDS

Local: bonsai-2:9b (Q2_0, 8192 ctx) + OpenRouter API

DESIGN STANDARDS

WCAG AA Contrast Compliant (14.2:1 Primary) · Offline-First

1. Brand Vision & Design Principles

HINAKU (ヒナク) is inspired by the Japanese phrase *Hikari naki uchū* (光なき宇宙), meaning *"a universe without light"*. The interface is purposefully built as a deep, calm, distraction-free desktop environment that feels quiet, contemplative, and relentlessly loyal to its single owner.

> **Core Design Motto:**
>
> *"Darkness, infinity, mystery, solitude. One name. An infinite void. No light at the end."*
>
> This identity governs atmospheric tone and typography only. It never manifests as pessimism or hesitation in replies; HINAKU responds with precision, dry wit, and decisive clarity.

Quiet, Modern Composure

Zero high-contrast glaring lines, zero promotional popups, and zero flashy animations. Surfaces follow soft neutral-dark elevations with one muted deep violet accent.

Local-First & Private

Zero external Google Fonts, CDNs, or tracking scripts. All SVGs and CSS are bundled locally on the machine. Personal memories and private documents are never sent across network boundaries.

Truthful Backend Attribution

Every message displays a clear badge indicating whether it was produced by the local GPU model, an external API, or safety guards, with complete execution latency telemetry.

Tactile Desktop Ergonomics

Designed specifically for an HP V15 laptop with full keyboard navigation (Ctrl+B, Ctrl+., Ctrl+N, Ctrl+K), global tray summon hotkey, and smooth resizability across 1280px, 960px, and 720px viewports.

2. Color Tokens & Surface System

HINAKU operates exclusively in a dark palette ("a universe without light"). There is no light mode. Surface depth is expressed through four primary elevation levels that provide visual hierarchy without harsh dividers.

Surface Elevation Palette

Base Canvas
`--bg-base: #141414`

Surface Viewport
`--bg-surface: #1f1f1f`

Raised Elements
`--bg-raised: #2a2a2a`

Overlay & Modals
`--bg-overlay: #333333`

Subtle Hover Fill
`rgba(255,255,255,0.04)`

Subtle Border
`rgba(255,255,255,0.08)`

Semantic & Status Color Tokens

TOKEN	HEX / VALUE	WCAG CONTRAST	SEMANTIC PURPOSE & USAGE
`--accent-hinaku`	`#8b5cf6` / `#7c3aed`	5.4:1 on #141414	Primary brand identity, active focus rings, send action buttons, prominent icons.
`--color-local`	`#3b82f6`	5.1:1 on #1f1f1f	Local execution indicator (Ollama bonsai-2:9b), local model tags.
`--color-online`	`#a855f7`	6.2:1 on #1f1f1f	Cloud API execution badge (OpenRouter/Claude/GPT), logged by provider warning.
`--color-private`	`#10b981`	7.8:1 on #1f1f1f	Private session banner, local-only confirmation, zero memory extraction badge.
`--color-warning`	`#f59e0b`	9.2:1 on #1f1f1f	Guard intervention notices, auto-private fallback alerts, hardware throttles.
`--color-error`	`#ef4444`	5.8:1 on #1f1f1f	Backend connection failures, invalid input errors, token budget overruns.

3. Typography & Spacing System

Text hierarchy is tuned for prolonged reading and dense technical documentation without visual fatigue. The primary font stack prioritizes Windows 11's native `Segoe UI Variable` for crisp rendering, paired with `Cascadia Code` for syntax-highlighted code.

Typography Scale

ROLE	FONT FAMILY	SIZE / LEADING	WEIGHT	USAGE CONTEXT
Display / H1	Segoe UI Variable Display	22px / 1.25	600 (Semibold)	Empty state greeting ("What are we working on?"), full-page headers.
Section / H2	Segoe UI Variable Text	18px / 1.30	600 (Semibold)	Document walkthrough Unit headers, settings category titles.
Topic / H3	Segoe UI Variable Text	16px / 1.40	600 (Semibold)	Topic titles in walkthroughs, card group headers.
Subheading / H4	Segoe UI Variable Text	14.5px / 1.50	500 (Medium)	Inner section headings (e.g. Definition, Breakdown, Takeaways).
Body Primary	Segoe UI Variable Text	14.5px / 1.60	400 (Regular)	Assistant replies, user messages, conversational dialogue.
Metadata / Small	Segoe UI Variable Text	12.5px / 1.50	400 / 500	Message timestamps, model badges, action chip labels.
Caption / Micro	Segoe UI Variable Text	11px / 1.40	500 (Medium)	Telemetry parameters, keyboard shortcut badges, status indicators.
Code Blocks	Cascadia Code, Consolas	13px / 1.55	400 (Regular)	Inline code snippets, multi-line Python/JS syntax blocks.

Spacing & Radii Scale

4px Spacing Scale

`--space-1`	4px	Micro gap between badges and icons
`--space-2`	8px	Card padding, button horizontal padding
`--space-3`	12px	Gap between message bubbles and meta rows
`--space-4`	16px	Container padding, drawer margins

Border Radii Scale

`--radius-sm`	6px	Chips, status badges, code inline tags
`--radius-md`	10px	User message bubbles, dropdown menus
`--radius-lg`	16px	Settings cards, document attachment drawers
`--radius-composer`	24px	Floating interaction composer dock

`--space-6`	24px	Stage padding, composer dock margin
`--space-8`	32px	Major section vertical separation

`--radius-full`	9999px	Action pills, toggle switches, status dots

4. Shell Architecture & Responsive Layout

The desktop shell implements a flexible 3-column architecture designed for native desktop performance on Windows 11. All transitions utilize `150ms ease-out` or `250ms cubic-bezier(0.16, 1, 0.3, 1)` for instantaneous, fluid resizing without layout thrashing.

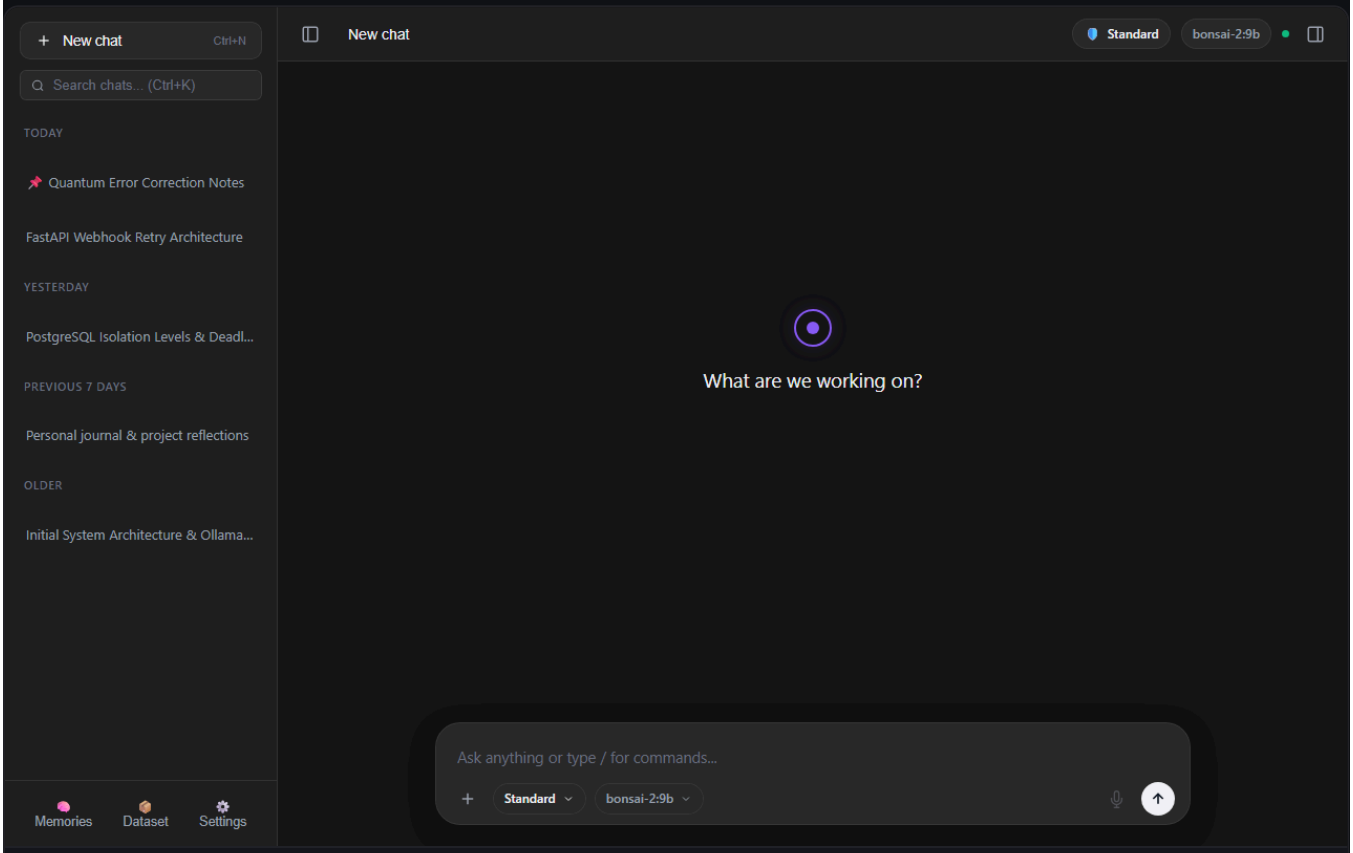


Figure 1: Complete 3-Column Desktop Shell (1280px Breakpoint) Sidebar (260px) + Conversation Stage (720px max) + Context Drawer (320px)

Responsive Breakpoint Matrix

BREAKPOINT	LEFT SIDEBAR (HISTORY)	CENTER STAGE (CONVERSATION)	RIGHT CONTEXT DRAWER
1280px+ (Full)	Expanded (260px fixed width). Full date categorization.	Centered 720px reading column with floating bottom composer.	Docked (320px fixed width). Real-time semantic memory inspector.
960px – 1279px (Compact)	Expanded (240px width). Auto-collapse via `Ctrl+B`.	Expanded to fit viewport; composer max-width 680px.	Auto-collapsed to icon tab in top nav; slides in as overlay on demand.
< 960px / 720px (Single-Col)	Collapsed by default. Slides in as overlay modal via top hamburger.	100% viewport width with 12px gutter. Composer bottom-pinned.	Accessible via full-page view or slide-in right drawer.

Top Navigation Bar Elements

The top bar (height: `48px`) provides unobtrusive environmental awareness:

- **Left:** Sidebar toggle icon (`Ctrl+B`) and editable conversation title.
- **Center:** Model status pill with pulsing green indicator dot (`bonsai-2:9b · 100% GPU`).
- **Right:** Privacy mode badge (`Standard` / `Local-Only` / `Private`), Context Drawer toggle (`Ctrl+.`), and Window controls.

5. Floating Composer & Interaction Dock

The prompt composer is engineered as a floating elevated pill (`--radius-composer: 24px`) positioned at the bottom of the conversation stage. It eliminates clutter while providing immediate access to model modes and attachments.

Auto-Expanding Textarea

Expands seamlessly from single-line (`44px`) to multi-line (up to `180px` max before scroll). Submits on `Enter`; inserts line break on `Shift + Enter`.

Dual-State Action Button

Circular submit button (`36px`) with deep violet fill (`#8b5cf6`). During active streaming, dynamically transforms into a Stop Generation square (■) with immediate abort capability.

Removable Mode Chips (Think Harder & Detailed Mode)

Clicking the `+` button opens a lightweight action menu that toggles specialized reasoning and depth flags:

MODE	BADGE DISPLAY	INFERENCE PARAMETER SHIFT	BEHAVIORAL EFFECT
Think Harder	🧠 THINK	`num_predict: 2048` , `temperature: 0.2`	Allocates deep token budget for complex logic, math, and code architecture problems.
Detailed Mode	📄 DETAILED ×	`num_predict: 800` , detailed persona override	Forces structured, comprehensive multi-paragraph explanations ending in a clear conclusion.

Privacy Mode Selector

STANDARD MODE

Intelligent hybrid routing. Simple conversational queries run on local `bonsai-2:9b` ; complex reasoning may route to external API models if configured.

LOCAL-ONLY MODE

Strict physical isolation. 100% of queries execute through local Ollama on the laptop's RTX 4050 GPU. Zero outbound internet traffic.

PRIVATE MODE

Confidential mode. Strictly local GPU execution. Suppresses all semantic memory extraction and excludes turns from the 6-month self-build dataset.

AUTO-PRIVATE FALLBACK

Automated safety guard. When passwords, PINs, or sensitive identifiers are typed, the system automatically switches to Private mode and displays a persistent banner with a 1-click **Undo** button.

6. Conversation View & Truthful Telemetry

Every conversation turn balances visual elegance with rigorous telemetry truthfulness. User inputs and assistant outputs are visually differentiated to maintain clear conversational momentum.

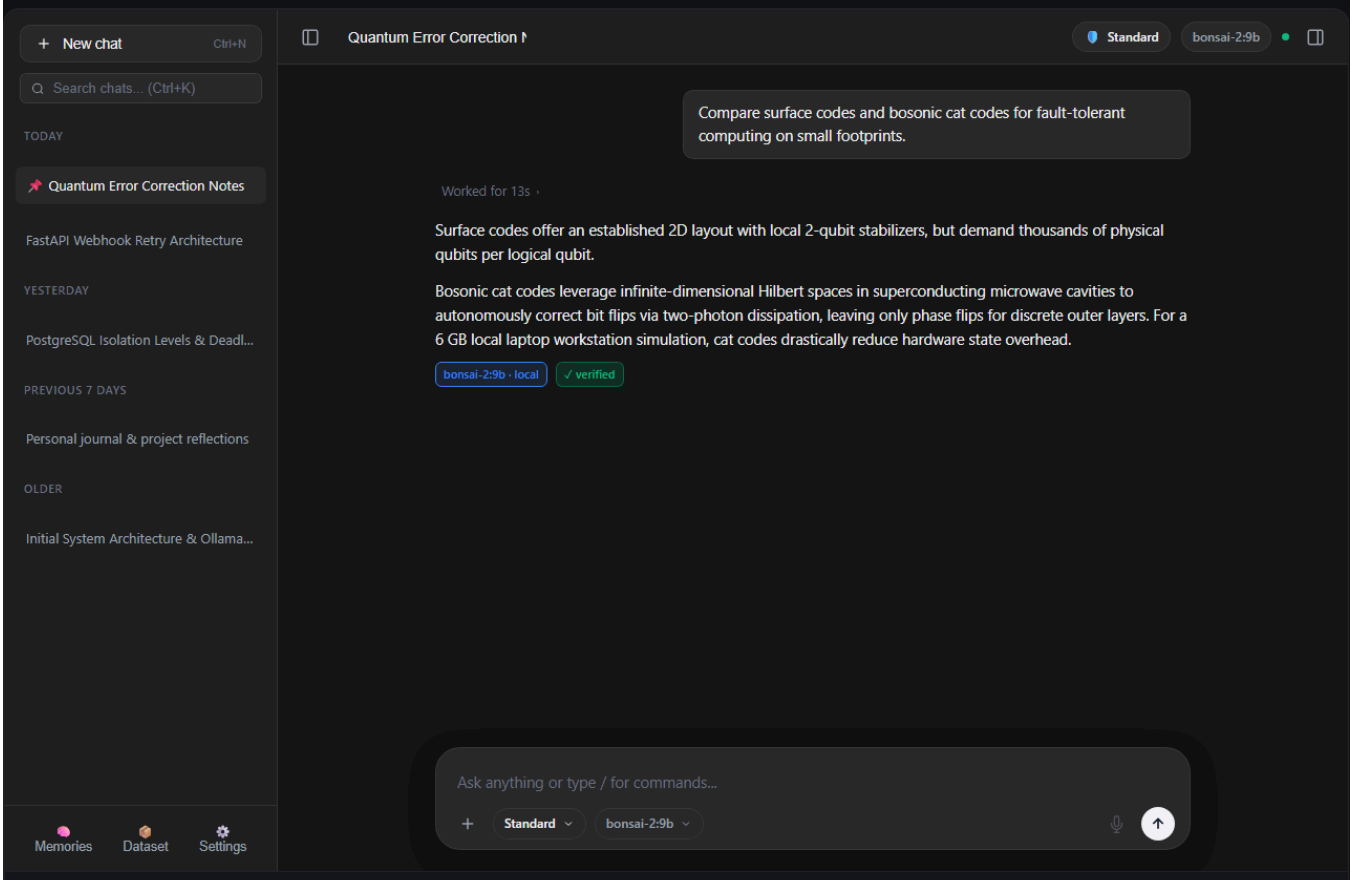


Figure 2: Active Conversation View *User message fold, markdown rendering, syntax code block, and telemetry accordion*

Key Message Components

User Message Bubble: Right-aligned elevated card (`--bg-raised: #2a2a2a`) with rounded corners (`10px`). Messages exceeding 350 characters automatically fold with a subtle gradient mask and a tactile *"Show more ↓"* toggle.

Assistant Message Body: Fluid left-aligned markdown typography. Headings (`###` , `####`) render as distinct section headers; unordered and numbered lists render with generous padding; horizontal rules (`---`) render as subtle dividers.

Syntax-Highlighted Code Blocks: Wrapped in dark pre-formatted containers (`#141416`) withCasdadia Code typography, a top header showing the language tag (e.g. `python` , `bash`), and a 1-click Copy button that displays *"Copied!"* confirmation.

Truthful Backend Badges: BONSAI-2:9B · LOCAL OPENROUTER · API · LOGGED BY PROVIDER ✓ VERIFIED

Telemetry Accordion ("Worked for X s >"): A collapsible process summary located under assistant replies. It surfaces exact execution timings (TTFT, generation latency, tokens per second, memory extraction time), *while strictly suppressing raw internal reasoning scratchpads* to preserve HINAKU's cool composure.

Message Hover Action Bar: Floating horizontal action dock appearing on message hover: **Copy Text**, **Thumbs Up**, **Thumbs Down**, **Edit Reply** (which feeds the 6-month curation engine), **Retry**, and **Continue** (for automated continuation of truncated replies).

7. Document Attachments & Walkthrough UX

Built during Phase 2 and refined in Phase 2c, the document attachment subsystem allows the owner to attach PDF, DOCX, TXT, and HTML study notes. The UI deterministically routes between overview study walkthroughs and targeted retrieval Q&A.

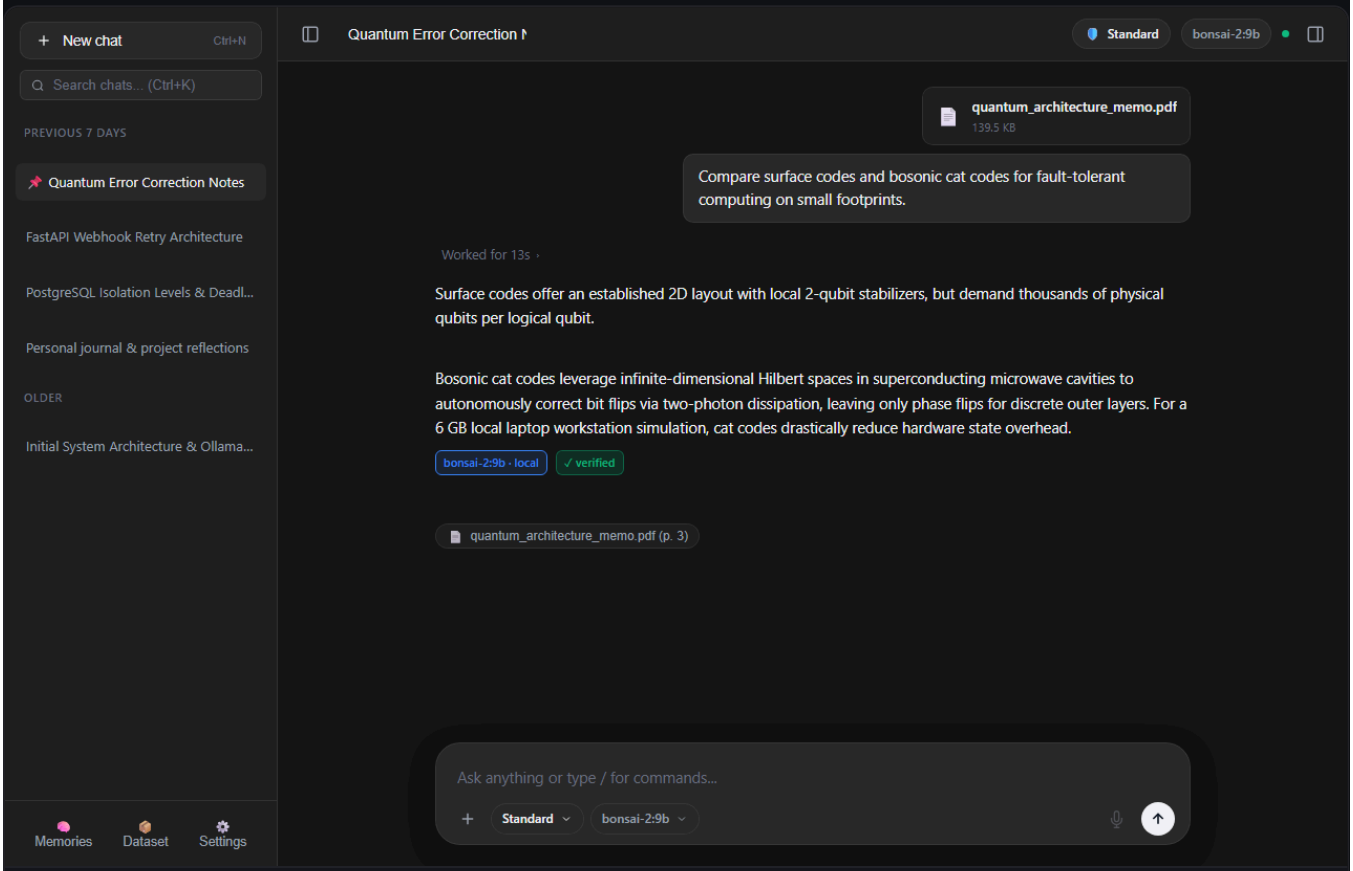


Figure 3: Document Attachment Drawer & Study Notes Integration *Uploaded document badge, outline section counts, and structured study cards*

Document Walkthrough Features

FEATURE	UI ELEMENT / BEHAVIOR	ENGINEERING SPECIFICATION
Attachment Drawer	Slide-down card above composer displaying filename, size (KB), and token count.	Supports drag-and-drop & file picker; validates disk headroom (file + 2 GB) before ingestion.
Outline Extraction	Extracts units, topics, sessions, study aids, and exact page ranges.	Stored in SQLite `attachment_sections`; indexed into isolated Chroma collection.
Intent Gating	Evaluates query in code: overview requests trigger walkthrough; specific questions trigger hybrid retrieval.	Filters out unrelated chatter (e.g. Python coding, CPUs vs GPUs) with zero false document citations.

FEATURE	UI ELEMENT / BEHAVIOR	ENGINEERING SPECIFICATION
Structure Preamble	Deterministic intro card: *"Your notes contain 2 units, 9 topics in Unit 1, 5 sessions in Unit 2..."*	Displays measured time estimate range: e.g. `~175s estimated (range 149s–201s)`.
Rolling Estimate	Progress banner updating topic-by-topic: `Explaining topic 3 of 9 (~120s remaining)`.	Derived from rolling average of completed topics; strictly non-increasing.
Action Chips	Structured button pill at the bottom of document replies: `Explain the whole document →`	Replaces raw markdown anchor links; triggers full walkthrough on click.
Code-Level Citations	Exact outline metadata note: `*Based on Session 8 — Introduction to MLSecOps (pp. 8–9).*`	Generated strictly in code from outline records; all model-hallucinated citations are stripped.

8. Dedicated Utility Pages

HINAKU eliminates intrusive modal popups in favor of dedicated, distraction-free utility views. These views share the same left sidebar navigation and allow full inspection of the system's memory, training data, and settings.

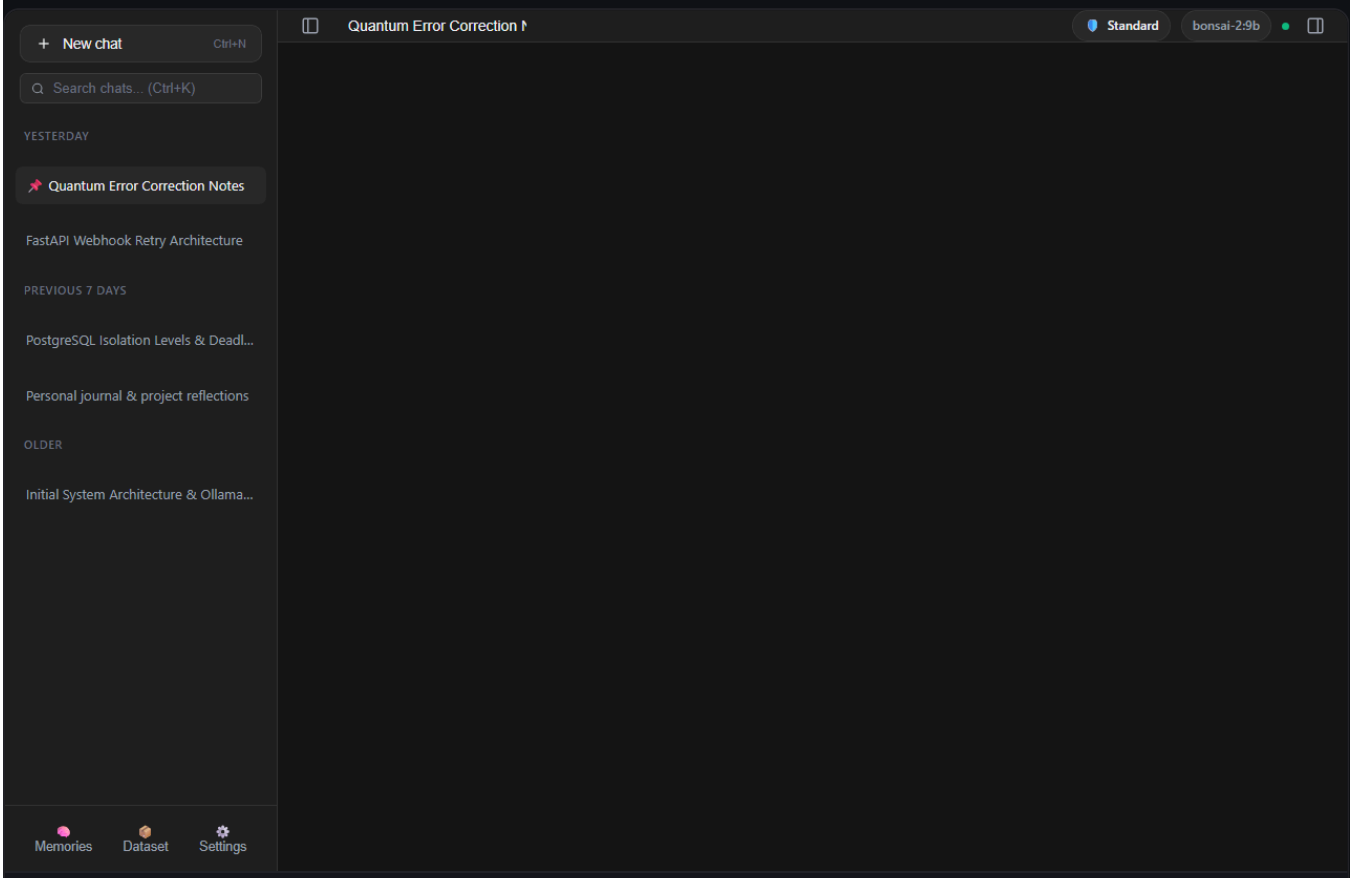


Figure 4: Context & Semantic Memory Management ViewSearchable memory cards, inline editing, category tagging, and re-index controls

1. Semantic Memory Drawer

Displays all durable facts and preferences extracted about the owner. Each memory card shows the extracted text, creation timestamp, and category badge. Users can search memories with semantic similarity, edit wording inline, delete obsolete facts, or trigger an on-demand ChromaDB re-index.

2. Dataset Curation Screen

Feeds the 6-month self-build cycle. Aggregates all interaction pairs flagged with thumbs-up or owner corrections. Provides review cards where the owner can audit training pairs, delete ungrounded entries, and view readiness against the minimum fine-tuning sample threshold.

3. Model Ratings & Bake-Off

Presents real-time telemetry comparing local models (`bonsai-2:9b` , `gemma3:4b`) against API models. Displays latency distributions,

4. Settings & Data Sovereignty

Configures default local model, router mode, global hotkeys, and reduced-motion toggles. Provides 1-click **Export All Data** (JSON format)

upvote/downvote ratios, token generation speeds, and persona adherence scores.

and a fail-safe **Permanent Wipe** (`WIPE_ALL`) that clears all SQLite databases, Chroma indices, and disk caches.

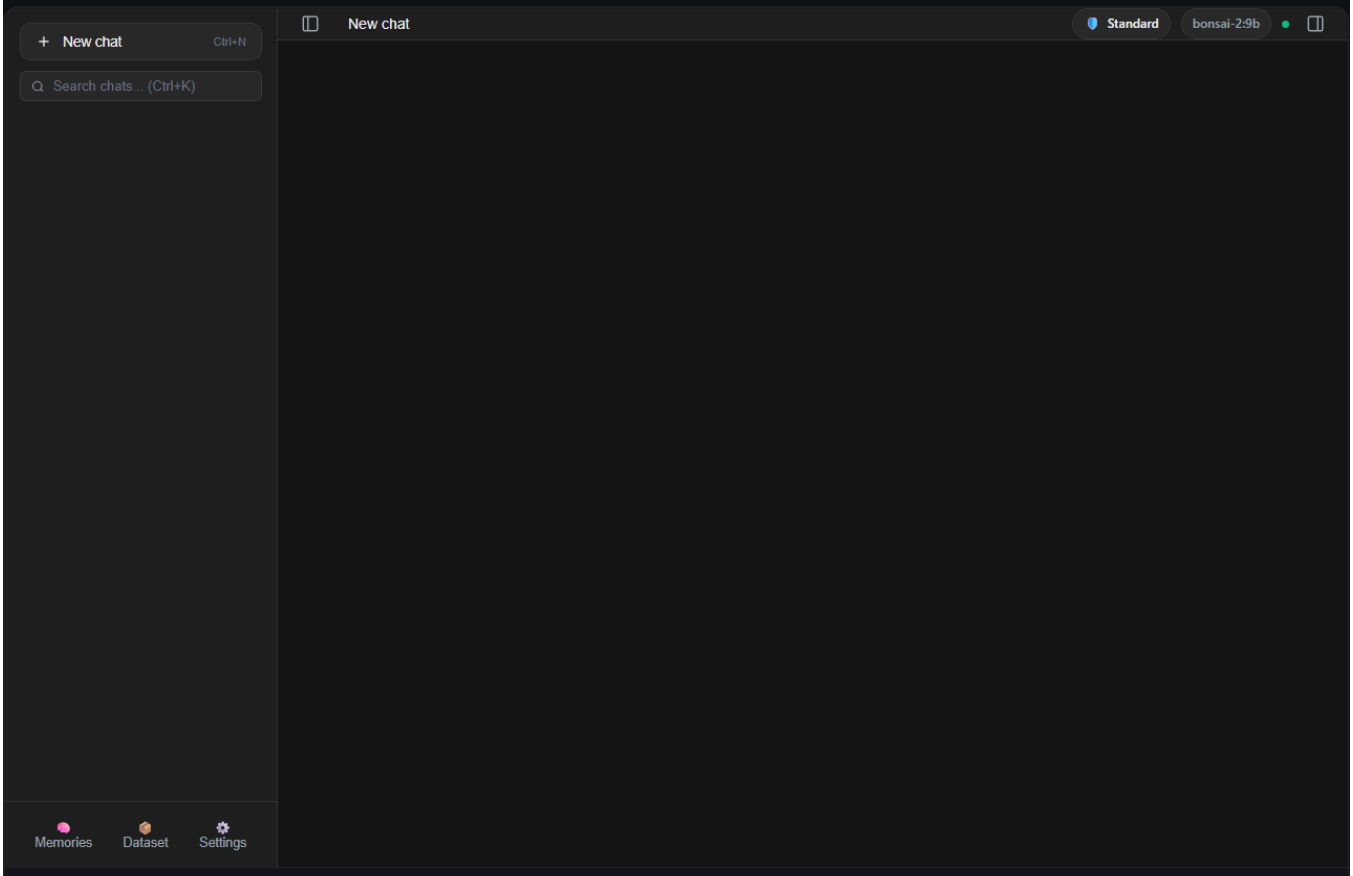


Figure 5: System Settings & Hardware Configuration *Router rules, memory extract toggles, global shortcuts, and export/wipe utilities*

9. Keybindings, Motion & Accessibility

Every feature in HINAKU is fully operable without a mouse. Micro-interactions are snappy and respect hardware efficiency and accessibility standards.

Complete Keyboard Shortcut Cheat Sheet

SHORTCUT	ACTION NAME	FUNCTIONAL DESCRIPTION
`Ctrl + N`	New Chat	Clears conversation viewport, resets composer, and initializes a clean conversation ID.
`Ctrl + K`	Search Chats	Focuses the search filter input in the left sidebar to quickly locate past conversation titles.
`Ctrl + B`	Toggle Sidebar	Collapses or expands the left conversation history sidebar with smooth 150ms ease transition.
`Ctrl + .`	Toggle Context Panel	Slides open or closes the right-hand context drawer showing live memories and telemetry.
`Ctrl + Shift + P`	Cycle Privacy Mode	Cycles through Standard → Local-Only → Private mode with instant status indicator feedback.
`Ctrl + Shift + Space`	Summon / Hide	Global OS hotkey managed by desktop shell to summon HINAKU from the background or minimize to tray.
`Esc`	Stop / Dismiss	Immediately aborts streaming model generation; closes active dropdown menus or utility pages.
`Enter`	Send Prompt	Submits current prompt to the model router.
`Shift + Enter`	Newline	Inserts a clean line break in the expanding composer textarea without submitting.

Motion & Accessibility Engineering

Reduced Motion Compliance

Strictly observes the OS-level `prefers-reduced-motion: reduce` media query. When enabled, all CSS transitions collapse from 150ms/250ms to `0ms`, and the subtle ambient star-field canvas is completely disabled to prevent disorientation.

Screen Reader & Focus Rings

All interactive controls feature a prominent `outline: 2px solid var(--accent-hinaku)` focus state. Streaming assistant responses emit into an `aria-live="polite"` region so screen readers receive updates without interruption.

HINAKU UI/UX Design System Specification · Compiled October 2026 · Built for HP V15 Target Environment